

□ SECTION A – BASIC (1–40)

1. Stress is defined as:

- A) Force
- B) Force per unit area
- C) Energy per unit volume
- D) Work done

2. First law of thermodynamics is based on conservation of:

- A) Mass
- B) Energy
- C) Momentum
- D) Heat

3. Reynolds number determines:

- A) Pressure
- B) Type of flow
- C) Density
- D) Work

4. Young's modulus equals:

- A) Stress/Strain
- B) Strain/Stress
- C) Load/Area
- D) Force $\times$ Length

5. Otto cycle is used in:

- A) Diesel engine
- B) Petrol engine
- C) Steam turbine
- D) Compressor

6. Heat transfer by conduction mainly occurs in:

- A) Solids
- B) Vacuum
- C) Space
- D) Plasma

7. Bending moment unit is:

- A) N
- B) N·m
- C) Pa
- D) J

8. Cavitation in pump occurs due to:

- A) High pressure
- B) Low pressure

- C) High viscosity
- D) High density

9. Steel is alloy of:

- A) Fe + Cu
- B) Fe + C
- C) Fe + Zn
- D) Fe + Al

10. Thermal efficiency is:

- A) Q_{in}/W_{out}
- B) W_{out}/Q_{in}
- C) $W \times Q$
- D) $Q - W$

11. Shear stress acts:

- A) Perpendicular
- B) Parallel
- C) Radial
- D) Circular

12. Laminar flow occurs when Re is:

- A) >4000
- B) <2000
- C) $=3000$
- D) >10000

13. Lathe machine performs:

- A) Turning
- B) Milling
- C) Shaping
- D) Broaching

14. Entropy measures:

- A) Energy
- B) Disorder
- C) Work
- D) Pressure

15. Buckling failure occurs in:

- A) Beams
- B) Columns
- C) Shafts
- D) Springs

16. Hydraulic turbine converts:

- A) Mechanical to hydraulic
- B) Hydraulic to mechanical
- C) Heat to work
- D) Electrical to mechanical

17. Annealing increases:

- A) Hardness
- B) Ductility
- C) Brittleness
- D) Residual stress

18. Gear ratio is ratio of:

- A) Torque
- B) Speed
- C) Load
- D) Area

19. Unit of viscosity is:

- A) Pa·s
- B) N
- C) m/s
- D) kg

20. Neutral axis in bending has:

- A) Max stress
- B) Zero stress
- C) Max strain
- D) Zero load

21. Enthalpy equals:

- A) $U - PV$
- B) $U + PV$
- C) PV
- D) $T + S$

22. Rolling process reduces:

- A) Thickness
- B) Length
- C) Hardness
- D) Density

23. Bearing reduces:

- A) Load
- B) Friction
- C) Speed
- D) Torque

24. Ideal gas equation is:

- A) $PV = mRT$
- B) $PV = RT$
- C) $P + V = RT$
- D) $PV = T$

25. Hardness is measured by:

- A) Brinell test

- B) Tension test
- C) Fatigue test
- D) Creep test

26. Pascal's law applies to:

- A) Solids
- B) Fluids
- C) Gases only
- D) Heat

27. Spring stores energy as:

- A) Kinetic
- B) Potential
- C) Thermal
- D) Chemical

28. Diesel cycle heat addition at constant:

- A) Volume
- B) Pressure
- C) Temperature
- D) Entropy

29. Convection requires:

- A) Solid
- B) Fluid
- C) Vacuum
- D) Metal only

30. Factor of safety =

- A) Yield stress / Working stress
- B) Working / Yield
- C) Load / Area
- D) Stress $\times$ Strain

31. CNC stands for:

- A) Computer Numerical Control
- B) Central Numeric Cutting
- C) Controlled Number Code
- D) Computer Normal Control

32. Flow becomes turbulent when Re is:

- A) <2000
- B) >4000
- C) $=3000$
- D) <1000

33. Corrosion is mainly due to:

- A) Oxidation
- B) Reduction

- C) Heating
- D) Hardening

34. Cp is specific heat at constant:

- A) Volume
- B) Pressure
- C) Temp
- D) Entropy

35. Work done in thermodynamic cycle equals area under:

- A) PV diagram
- B) TS diagram
- C) PT diagram
- D) TV diagram

36. Milling cutter is:

- A) Single point
- B) Multi-point
- C) No point
- D) Double point

37. Density =

- A) Mass/Volume
- B) Force/Area
- C) Work/Time
- D) Energy/Volume

38. Reversible process is:

- A) Frictionless
- B) Fast
- C) Irreversible
- D) Real only

39. Kinematic pair transmits:

- A) Motion
- B) Heat
- C) Pressure
- D) Mass

40. Heat exchanger transfers:

- A) Work
- B) Heat
- C) Mass
- D) Pressure

☐ **SECTION B – MEDIUM (41–80)**

41. Maximum bending stress in beam occurs at:

- A) Neutral axis
- B) Extreme fibers

- C) Mid depth
- D) Centroid

42. Bernoulli equation is derived from conservation of:

- A) Mass
- B) Energy
- C) Momentum
- D) Entropy

43. LMTD is used in design of:

- A) Boilers
- B) Heat exchangers
- C) Turbines
- D) Pumps

44. Torsion equation is:

- A) $M/I = \sigma/y$
- B) $T/J = \tau/r$
- C) F/A
- D) $PV = mRT$

45. Carnot efficiency depends on:

- A) Pressure
- B) Volume
- C) Temperature limits
- D) Working fluid

46. Critical radius of insulation depends on:

- A) k & h
- B) Density
- C) Pressure
- D) Volume

47. Centrifugal pump works on:

- A) Positive displacement
- B) Centrifugal force
- C) Impact
- D) Compression

48. Fatigue failure occurs due to:

- A) Static load
- B) Cyclic load
- C) Impact
- D) Temperature

49. Brinell hardness test uses:

- A) Steel ball
- B) Diamond cone
- C) Needle
- D) Hammer

50. Governing in engine controls:

- A) Pressure
- B) Speed
- C) Torque
- D) Temperature

51. Euler's formula is used to determine:

- A) Bending stress in beams
- B) Buckling load of long columns
- C) Torsional stress in shafts
- D) Shear stress in rivets

52. Volumetric efficiency is an important parameter in:

- A) Steam turbine
- B) IC engine
- C) Centrifugal pump
- D) Heat exchanger

53. Poisson's ratio is defined as:

- A) Longitudinal strain / Lateral strain
- B) Lateral strain / Longitudinal strain
- C) Shear stress / Shear strain
- D) Stress / Strain

54. Compared to parallel flow, a counter-flow heat exchanger generally has:

- A) Lower efficiency
- B) Same efficiency
- C) Higher efficiency
- D) Zero effectiveness

55. Proper machining primarily improves:

- A) Material hardness
- B) Surface finish
- C) Density
- D) Chemical composition

56. A hydraulic press operates based on:

- A) Bernoulli's principle
- B) Pascal's law
- C) Fourier's law
- D) Euler's equation

57. The degree of freedom of a rigid body in plane motion is:

- A) 1
- B) 2
- C) 3
- D) 6

58. Specific speed is used to classify:

- A) Boilers
- B) Pumps and turbines
- C) Heat exchangers
- D) Compressors

59. Modulus of rigidity (G) is the ratio of:

- A) Normal stress / Normal strain
- B) Shear stress / Shear strain
- C) Bending stress / Strain
- D) Torsion / Twist

60. A phase diagram shows:

- A) Stress–strain relationship
- B) Temperature–composition relationship
- C) Pressure–volume relationship
- D) Load–deflection curve

61. Stress concentration factor is defined as the ratio of:

- A) Average stress / Maximum stress
- B) Maximum stress / Nominal stress
- C) Nominal stress / Yield stress
- D) Working stress / Ultimate stress

62. The efficiency of the Otto cycle depends mainly on:

- A) Cut-off ratio
- B) Compression ratio
- C) Maximum pressure
- D) Engine speed

63. Head loss due to friction in a pipe is calculated using:

- A) Bernoulli equation
- B) Darcy–Weisbach equation

- C) Euler equation
- D) Laplace equation

64. The main components of the Rankine cycle include:

- A) Compressor, turbine, condenser
- B) Boiler, turbine, condenser, pump
- C) Engine, pump, boiler
- D) Turbine and pump only

65. Tempering after quenching is done to:

- A) Increase brittleness
- B) Reduce brittleness and improve toughness
- C) Increase hardness further
- D) Remove corrosion

66. Gear interference occurs when:

- A) Teeth are properly meshed
- B) Involute portion is not fully engaged
- C) Excess lubrication is applied
- D) Gear ratio is unity

67. The primary function of a clutch is to:

- A) Increase torque
- B) Connect and disconnect power transmission
- C) Reduce friction
- D) Change speed

68. Break-even point occurs when:

- A) Profit is maximum
- B) Revenue equals total cost
- C) Variable cost is zero
- D) Fixed cost is maximum

69. The basic EOQ formula balances:

- A) Production and sales
- B) Ordering cost and holding cost
- C) Labor and material cost
- D) Fixed and variable cost

70. Boundary layer separation is mainly caused by:

- A) Favorable pressure gradient
- B) Adverse pressure gradient
- C) High density
- D) Low viscosity

71. The cut-off ratio in Diesel cycle is the ratio of:

- A) Compression ratio
- B) Volumes before and after heat addition
- C) Pressures
- D) Temperatures

72. In a simply supported beam with central load, maximum deflection occurs at:

- A) Supports
- B) Mid-span
- C) Quarter span
- D) Fixed end

73. The purpose of a flywheel in engines is to:

- A) Increase power
- B) Store excess energy and reduce speed fluctuation
- C) Increase torque permanently
- D) Reduce fuel consumption

74. Specific fuel consumption of an engine is defined as:

- A) Fuel used per unit time
- B) Fuel used per unit power output
- C) Fuel used per cycle
- D) Fuel per unit volume

75. In heat transfer through fins, fins are provided to:

- A) Reduce surface area
- B) Increase heat dissipation area
- C) Increase temperature
- D) Reduce conductivity

76. The maximum shear stress theory is also known as:

- A) Rankine theory
- B) Tresca theory

- C) Von Mises theory
- D) Coulomb theory

77. In a centrifugal pump, priming is required to:

- A) Increase speed
- B) Remove air from casing
- C) Increase pressure
- D) Reduce cavitation

78. The moment of inertia of a section affects:

- A) Axial stress
- B) Bending stress
- C) Shear stress only
- D) Density

79. Heat exchanger effectiveness is defined as:

- A) Actual heat transfer / Maximum possible heat transfer
- B) Q_{in} / Q_{out}
- C) Temperature ratio
- D) Heat loss / Heat gain

80. In a four-stroke engine, one power stroke occurs in:

- A) One revolution
- B) Two revolutions
- C) Half revolution
- D) Four revolutions

☐ **SECTION C – HARD (81–100)**

81. Otto cycle efficiency depends on:

- A) Compression ratio only
- B) Cut-off ratio
- C) Temperature
- D) Pressure

82. Maximum shear stress theory is given by:

- A) Rankine
- B) Tresca
- C) von Mises
- D) Coulomb

83. In thick cylinder, stress distribution is given by:

- A) Euler theory
- B) Lamé's equation
- C) Bernoulli
- D) Fourier

84. Boundary layer thickness increases with:

- A) Velocity
- B) Distance from leading edge
- C) Pressure
- D) Density

85. Heat transfer in fins is mainly to:

- A) Increase area
- B) Increase k
- C) Increase temp
- D) Reduce heat

86. Degree of freedom (Gruebler's equation) for mechanism:

- A) $3(n-1)-2j$
- B) $n-2j$
- C) $2n-j$
- D) $n+j$

87. In Rankine cycle, pump work is:

- A) Negligible
- B) Maximum
- C) Zero
- D) Equal to turbine

88. Stress concentration reduced by:

- A) Sharp corner
- B) Fillet
- C) Hole
- D) Notch

89. Specific speed of turbine indicates:

- A) Type
- B) Power
- C) Torque
- D) Work

90. Fourier law relates heat flux to:

- A) Temperature gradient
- B) Pressure
- C) Volume
- D) Entropy

91. The Brayton cycle is ideally used in:

- A) Steam engines
- B) Gas turbines
- C) Diesel engines
- D) Hydraulic turbines

92. Mohr's circle is primarily used to determine:

- A) Heat transfer rate
- B) Stress transformation
- C) Flow velocity
- D) Hardness

93. In a centrifugal governor, the concept analogous to slip in an induction motor refers to:

- A) Speed regulation difference
- B) Load variation
- C) Torque fluctuation
- D) Power loss

94. Creep deformation in materials becomes significant at:

- A) Low temperature
- B) Room temperature
- C) High temperature
- D) Zero stress

95. The S–N curve represents the relationship between:

- A) Stress and strain
- B) Stress and fatigue life
- C) Stress and temperature
- D) Speed and torque

96. Head loss due to friction in pipes is calculated using:

- A) Bernoulli equation
- B) Fourier law
- C) Darcy–Weisbach equation
- D) Euler equation

97. The effectiveness–NTU method in heat exchanger analysis uses NTU which stands for:

- A) Number of Thermal Units
- B) Net Temperature Unit
- C) Number of Transfer Units
- D) Normal Thermal Usage

98. A clutch transmits torque mainly through:

- A) Hydraulic pressure
- B) Magnetic field
- C) Friction between contact surfaces
- D) Gear engagement

99. The critical speed of a rotating shaft occurs due to:

- A) Excess lubrication
- B) Resonance condition
- C) Overheating
- D) Gear failure

100. Entropy generation in a thermodynamic process is an indicator of:

- A) Energy conservation
- B) Reversibility
- C) Irreversibility
- D) Work output

✓ANSWER KEY

□ SECTION A – BASIC (1–40)

- 1. B
- 2. B
- 3. B
- 4. A
- 5. B
- 6. A
- 7. B
- 8. B
- 9. B
- 10. B
- 11. B
- 12. B
- 13. A
- 14. B
- 15. B
- 16. B
- 17. B
- 18. B

- 19. **A**
- 20. **B**
- 21. **B**
- 22. **A**
- 23. **B**
- 24. **A**
- 25. **A**
- 26. **B**
- 27. **B**
- 28. **B**
- 29. **B**
- 30. **A**
- 31. **A**
- 32. **B**
- 33. **A**
- 34. **B**
- 35. **A**
- 36. **B**
- 37. **A**
- 38. **A**
- 39. **A**
- 40. **B**

☐ **SECTION B – MEDIUM (41–80)**

- 41. **B**
- 42. **B**
- 43. **B**
- 44. **B**
- 45. **C**
- 46. **A**
- 47. **B**
- 48. **B**

- 49. **A**
- 50. **B**
- 51. **B**
- 52. **B**
- 53. **B**
- 54. **C**
- 55. **B**
- 56. **B**
- 57. **C**
- 58. **B**
- 59. **B**
- 60. **B**
- 61. **B**
- 62. **B**
- 63. **B**
- 64. **B**
- 65. **B**
- 66. **B**
- 67. **B**
- 68. **B**
- 69. **B**
- 70. **B**
- 71. **B**
- 72. **B**
- 73. **B**
- 74. **B**
- 75. **B**
- 76. **B**
- 77. **B**
- 78. **B**
- 79. **A**
- 80. **B**

□ SECTION C – HARD (81–100)

81. **A**

82. **B**

83. **B**

84. **B**

85. **A**

86. **A**

87. **A**

88. **B**

89. **A**

90. **A**

91. **B**

92. **B**

93. **A**

94. **C**

95. **B**

96. **C**

97. **C**

98. **C**

99. **B**

100. **C**